

Unified Cox Lasso Algorithms

Fixed- s computation, cross-validation, and pre-validation

This document uses one common visual style for three Cox lasso algorithms: Tibshirani's fixed- s Cox lasso computation, Section 3.5.1 cross-validation, and Section 3.5.2 pre-validation.

Contents

1	Common notation	2
2	Fixed-s Cox lasso computation	2
2.1	Working quantities	2
2.2	Implementation notes for fixed- s computation	5
3	Cross-validation for the Cox lasso	6
3.1	Optional one-standard-error rule	6
3.2	Implementation notes for cross-validation	7
4	Pre-validation for the Cox lasso	7
4.1	Analysis of pre-validated scores	9
4.2	Optional final model for deployment	9
4.3	Implementation notes for pre-validation	9

1 Common notation

For right-censored survival data

$$(y_i, \delta_i, \mathbf{x}_i), \quad i = 1, \dots, N,$$

let y_i be the observed time, $\delta_i = 1$ indicate an observed failure, $\delta_i = 0$ indicate censoring, and $\mathbf{x}_i \in \mathbb{R}^p$ be the covariate vector. Let

$$\mathcal{N} = \{1, \dots, N\}.$$

Standing assumptions

Unless otherwise stated, assume independent, non-informative right censoring and no tied failure times. If tied failure times are present, use the same tie-handling convention, such as Breslow, Efron, or exact partial likelihood, consistently in all Cox partial likelihood evaluations.

For any subset $A \subseteq \mathcal{N}$, define

$$\mathcal{E}_A = \{i \in A : \delta_i = 1\}, \quad \mathcal{R}_i^A = \{j \in A : y_j \geq y_i\}, \quad i \in \mathcal{E}_A.$$

The Cox partial log-likelihood on subset A is

$$\ell_A(\boldsymbol{\beta}) = \sum_{i \in \mathcal{E}_A} \left[\mathbf{x}_i^\top \boldsymbol{\beta} - \log \left\{ \sum_{j \in \mathcal{R}_i^A} \exp(\mathbf{x}_j^\top \boldsymbol{\beta}) \right\} \right].$$

For the full data, write

$$\ell(\boldsymbol{\beta}) = \ell_{\mathcal{N}}(\boldsymbol{\beta}), \quad \text{Dev}_A(\boldsymbol{\beta}) = -2\ell_A(\boldsymbol{\beta}), \quad \text{Dev}(\boldsymbol{\beta}) = \text{Dev}_{\mathcal{N}}(\boldsymbol{\beta}).$$

2 Fixed- s Cox lasso computation

For fixed $s \geq 0$, the constrained Cox lasso estimator is

$$\hat{\boldsymbol{\beta}}(s) = \arg \min_{\boldsymbol{\beta} \in \mathbb{R}^p} \{-\ell(\boldsymbol{\beta})\} \quad \text{subject to} \quad \sum_{j=1}^p |\beta_j| \leq s.$$

Equivalently, one may maximize $\ell(\boldsymbol{\beta})$ subject to $\|\boldsymbol{\beta}\|_1 \leq s$.

2.1 Working quantities

For a current coefficient vector $\boldsymbol{\beta}$, define

$$\boldsymbol{\eta} = \mathbf{X}\boldsymbol{\beta}, \quad \eta_i = \mathbf{x}_i^\top \boldsymbol{\beta}.$$

For each $i \in \mathcal{E}$ and $k = 1, \dots, N$, where $\mathcal{E} = \mathcal{E}_{\mathcal{N}}$ and $\mathcal{R}_i = \mathcal{R}_i^{\mathcal{N}}$, define

$$\pi_{ik}(\boldsymbol{\eta}) = \frac{\mathbf{1}(k \in \mathcal{R}_i) \exp(\eta_k)}{\sum_{r \in \mathcal{R}_i} \exp(\eta_r)}.$$

The score with respect to $\boldsymbol{\eta}$ has components

$$u_k(\boldsymbol{\eta}) = \delta_k - \sum_{i \in \mathcal{E}} \pi_{ik}(\boldsymbol{\eta}), \quad k = 1, \dots, N.$$

The diagonal approximation to the observed information with respect to $\boldsymbol{\eta}$ is

$$D(\boldsymbol{\eta}) = \text{diag}\{d_1(\boldsymbol{\eta}), \dots, d_N(\boldsymbol{\eta})\},$$

where

$$d_k(\boldsymbol{\eta}) = \sum_{i \in \mathcal{E}} \pi_{ik}(\boldsymbol{\eta}) \{1 - \pi_{ik}(\boldsymbol{\eta})\}.$$

The IRLS pseudo-response is

$$z(\boldsymbol{\eta}) = \boldsymbol{\eta} + D(\boldsymbol{\eta})^\dagger u(\boldsymbol{\eta}),$$

where $D^\dagger$ denotes the generalized inverse of the diagonal matrix D .

Standardization convention

For the fixed- s algorithm, use standardized covariates and no intercept column in $\mathbf{X}$. If $x_{ij}^{\text{std}} = (x_{ij}^{\text{orig}} - \bar{x}_j)/\sigma_j$, then the original-scale coefficient is $\hat{\beta}_j^{\text{orig}} = \hat{\beta}_j^{\text{std}}/\sigma_j$. The Cox intercept is absorbed into the baseline hazard and is not separately identifiable.

Algorithm 1. CoxLassoFit($s, \varepsilon_{\text{conv}}, \tau_{\text{active}}, M$): fixed- s Cox lasso fit

Input. Survival data $\{(y_i, \delta_i, \mathbf{x}_i), i = 1, \dots, N\}$; standardized design matrix $\mathbf{X}$ with no intercept column; constraint parameter $s \geq 0$; convergence tolerance $\varepsilon_{\text{conv}} > 0$; active-set threshold $\tau_{\text{active}} > 0$; maximum iterations $M \geq 1$.

Output.

$$\hat{\boldsymbol{\beta}}(s), \quad \ell_s, \quad D_s, \quad z_s, \quad \lambda_s, \quad \hat{S}(s), \quad \text{status}.$$

1. Define $\mathcal{E} \leftarrow \{i : \delta_i = 1\}$ and $\mathcal{R}_i \leftarrow \{j : y_j \geq y_i\}$ for $i \in \mathcal{E}$.

2. Initialize

$$\boldsymbol{\beta}^{(0)} \leftarrow \mathbf{0}, \quad m \leftarrow 0, \quad \text{status} \leftarrow \text{NotConverged},$$

and set

$$D_{\text{last}} \leftarrow \text{NA}, \quad z_{\text{last}} \leftarrow \text{NA}, \quad \lambda_{\text{last}} \leftarrow \text{NA}.$$

3. If $s = 0$, set

$$\hat{\boldsymbol{\beta}}(s) \leftarrow \mathbf{0}, \quad \ell_s \leftarrow \ell(\mathbf{0}), \quad \hat{S}(s) \leftarrow \emptyset.$$

Compute $\boldsymbol{\eta} = \mathbf{X}\mathbf{0}$, set

$$D_s \leftarrow D(\boldsymbol{\eta}), \quad z_s \leftarrow z(\boldsymbol{\eta}), \quad \lambda_s \leftarrow \text{NA}, \quad \text{status} \leftarrow \text{Converged},$$

and return the output.

4. While $m < M$ and status = NotConverged, repeat:

(a) Compute $\boldsymbol{\eta}^{(m)} \leftarrow \mathbf{X}\boldsymbol{\beta}^{(m)}$.

(b) Compute $\pi_{ik}^{(m)}$, $u^{(m)}$, $D_w^{(m)}$, and $z^{(m)}$ using the working-quantity formulas above.

(c) Solve the constrained weighted least squares problem

$$\boldsymbol{\beta}^{\text{new}} \leftarrow \arg \min_{\boldsymbol{\beta} \in \mathbb{R}^p} \left\{ (z^{(m)} - \mathbf{X}\boldsymbol{\beta})^\top D_w^{(m)} (z^{(m)} - \mathbf{X}\boldsymbol{\beta}) \right\}$$

subject to

$$\sum_{j=1}^p |\beta_j| \leq s.$$

(d) Let λ^{new} be the Lagrange multiplier for the constraint, under the convention

$$\hat{\boldsymbol{\beta}}_{\mathcal{A}} \approx (\mathbf{X}_{\mathcal{A}}^\top D \mathbf{X}_{\mathcal{A}} + \lambda W_{\mathcal{A}})^\dagger \mathbf{X}_{\mathcal{A}}^\top D z.$$

If the constraint is inactive, set $\lambda^{\text{new}} = 0$.

(e) Store

$$D_{\text{last}} \leftarrow D_w^{(m)}, \quad z_{\text{last}} \leftarrow z^{(m)}, \quad \lambda_{\text{last}} \leftarrow \lambda^{\text{new}}.$$

(f) If

$$\|\beta^{\text{new}} - \beta^{(m)}\|_{\infty} < \varepsilon_{\text{conv}},$$

set status $\leftarrow$ Converged.

(g) Update $\beta^{(m+1)} \leftarrow \beta^{\text{new}}$ and $m \leftarrow m + 1$.

5. If status $\neq$ Converged, set status $\leftarrow$ MaxIterationReached.

6. Set

$$\begin{aligned} \hat{\beta}(s) &\leftarrow \beta^{(m)}, & \ell_s &\leftarrow \ell\{\hat{\beta}(s)\}, \\ D_s &\leftarrow D_{\text{last}}, & z_s &\leftarrow z_{\text{last}}, & \lambda_s &\leftarrow \lambda_{\text{last}}, \end{aligned}$$

and

$$\hat{S}(s) \leftarrow \{j : |\hat{\beta}_j(s)| > \tau_{\text{active}}\}.$$

Return the output.

Returned working quantities

The returned D_s, z_s, λ_s are from the final constrained weighted least squares subproblem. At convergence, they approximate the corresponding working quantities evaluated at $\hat{\beta}(s)$. If status = MaxIterationReached, the returned quantities are diagnostic only and should not be used for GCV or standard-error approximation.

Algorithm 2. GCVSelect($U, \varepsilon_{\text{conv}}, \tau_{\text{active}}, M$): select s by approximate GCV

Input. Survival data; standardized design matrix $\mathbf{X}$ with no intercept column; candidate grid $U = \{u_1, \dots, u_G\} \subset [0, 1]$; convergence tolerance $\varepsilon_{\text{conv}} > 0$; active-set threshold $\tau_{\text{active}} > 0$; maximum iterations $M \geq 1$.

Output.

$$\hat{s}_{\text{GCV}}, \hat{u}_{\text{GCV}}, \hat{\beta}_{\text{GCV}}, \ell_{\text{GCV}}, D_{\text{GCV}}, z_{\text{GCV}}, \lambda_{\text{GCV}}, \hat{S}_{\text{GCV}}, \text{status}.$$

1. Fit the ordinary unconstrained Cox model and obtain $\hat{\beta}^0$. Define

$$s_{\text{max}} \leftarrow \sum_{j=1}^p |\hat{\beta}_j^0|.$$

If the ordinary Cox estimate does not exist or is numerically unstable, choose a pre-specified upper bound s_{max} .

2. Set $\mathcal{V} \leftarrow \emptyset$, status $\leftarrow$ NotSelected, and choose a small numerical guard $\eta_{\text{gcv}} > 0$, for example 10^{-12} .

3. For $g = 1, \dots, G$, repeat:

(a) Set $s_g \leftarrow u_g s_{\text{max}}$.

(b) Run Algorithm 1:

$$(\hat{\beta}_g, \ell_g, D_g, z_g, \lambda_g, \hat{S}_g, \text{status}_g) \leftarrow \text{COXLASSOFIT}(s_g, \varepsilon_{\text{conv}}, \tau_{\text{active}}, M).$$

(c) If $\text{status}_g \neq$ Converged, set $\text{GCV}_g \leftarrow +\infty$ and continue to the next grid value.

(d) Define $\mathcal{A}_g \leftarrow \{j : |\hat{\beta}_{g,j}| > \tau_{\text{active}}\}$.

(e) If $\mathcal{A}_g = \emptyset$, set $p_g \leftarrow 0$. Otherwise set

$$\begin{aligned} W_{g,\mathcal{A}} &\leftarrow \text{diag}\{1/|\hat{\beta}_{g,j}| : j \in \mathcal{A}_g\}, \\ M_g &\leftarrow (\mathbf{X}_{\mathcal{A}_g}^\top D_g \mathbf{X}_{\mathcal{A}_g} + \lambda_g W_{g,\mathcal{A}})^\dagger \mathbf{X}_{\mathcal{A}_g}^\top D_g, \\ H_g &\leftarrow \mathbf{X}_{\mathcal{A}_g} M_g, & p_g &\leftarrow \text{tr}(H_g). \end{aligned}$$

(f) If $1 - p_g/N \leq \eta_{\text{gcv}}$, set $\text{GCV}_g \leftarrow +\infty$ and continue.

(g) Compute

$$\text{GCV}_g \leftarrow \frac{-\ell_g/N}{(1 - p_g/N)^2}.$$

(h) Add g to $\mathcal{V}$.

4. If $\mathcal{V} = \emptyset$, return NA, ..., NA, NotSelected.

5. Choose

$$g^* \leftarrow \arg \min_{g \in \mathcal{V}} \text{GCV}_g.$$

Return the selected fit:

$$\hat{u}_{\text{GCV}} \leftarrow u_{g^*}, \quad \hat{s}_{\text{GCV}} \leftarrow s_{g^*}, \quad \hat{\beta}_{\text{GCV}} \leftarrow \hat{\beta}_{g^*},$$

together with $\ell_{g^*}, D_{g^*}, z_{g^*}, \lambda_{g^*}, \hat{S}_{g^*}$, and status $\leftarrow$ Selected.

Algorithm 3. ApproxSE: approximate standard errors for the selected Cox lasso fit

Input. Standardized design matrix $\mathbf{X}$ with no intercept column; selected estimate $\hat{\beta}_{\text{GCV}}$; selected IRLS weight matrix D_{GCV} ; selected multiplier λ_{GCV} ; active-set threshold $\tau_{\text{active}} > 0$.

Output.

$$\widehat{\text{SE}}(\hat{\beta}_1), \dots, \widehat{\text{SE}}(\hat{\beta}_p), \quad \text{status}.$$

1. Set $\widehat{\text{SE}}(\hat{\beta}_j) \leftarrow \text{NA}$ for $j = 1, \dots, p$.

2. Define

$$\mathcal{A} \leftarrow \{j : |\hat{\beta}_{\text{GCV},j}| > \tau_{\text{active}}\}.$$

3. If $\mathcal{A} = \emptyset$, set status $\leftarrow$ Computed and return.

4. Form $\mathbf{X}_{\mathcal{A}}$ and

$$\mathbf{W}_{\mathcal{A}} \leftarrow \text{diag}\{1/|\hat{\beta}_{\text{GCV},j}| : j \in \mathcal{A}\}.$$

5. Compute

$$\mathbf{M}_{\mathcal{A}} \leftarrow (\mathbf{X}_{\mathcal{A}}^{\top} D_{\text{GCV}} \mathbf{X}_{\mathcal{A}} + \lambda_{\text{GCV}} \mathbf{W}_{\mathcal{A}})^{\dagger} \mathbf{X}_{\mathcal{A}}^{\top} D_{\text{GCV}},$$

and

$$\mathbf{V}_{\mathcal{A}} \leftarrow \mathbf{M}_{\mathcal{A}} D_{\text{GCV}}^{\dagger} \mathbf{M}_{\mathcal{A}}^{\top}.$$

6. For each $j \in \mathcal{A}$, let $r(j)$ be the position of j in the ordered active set and set

$$\widehat{\text{SE}}(\hat{\beta}_j) \leftarrow \sqrt{[\mathbf{V}_{\mathcal{A}}]_{r(j),r(j)}}.$$

7. Set status $\leftarrow$ Computed and return.

2.2 Implementation notes for fixed- s computation

Note 1. The constrained weighted least squares step can be solved as a quadratic program by writing $\beta_j = \beta_j^+ - \beta_j^-$, with $\beta_j^+, \beta_j^- \geq 0$, and imposing $\sum_j (\beta_j^+ + \beta_j^-) \leq s$.

Note 2. If the constraint is inactive, set $\lambda = 0$. If $s = 0$, the active set is empty and the GCV degrees of freedom are set to zero.

Note 3. If some diagonal entries of D are zero, use $D^{\dagger}$ or remove zero-weight observations in the corresponding weighted least squares calculation.

Note 4. Inactive variables are reported with standard errors NA, not as variables with true standard error zero.

Note 5. The convergence tolerance $\varepsilon_{\text{conv}}$ and active-set threshold τ_{active} are intentionally separated.

3 Cross-validation for the Cox lasso

For a fixed penalty $\lambda \geq 0$, define the penalized Cox lasso estimate on subset A by

$$\hat{\beta}_A(\lambda) = \arg \min_{\beta \in \mathbb{R}^p} \{-\ell_A(\beta) + \lambda \|\beta\|_1\}.$$

Algorithm 4. CoxLassoCV: K -fold cross-validation for the Cox lasso

Input. Survival data $\{(y_i, \delta_i, \mathbf{x}_i), i = 1, \dots, N\}$; penalty grid $\Lambda = \{\lambda_1, \dots, \lambda_M\}$; number of folds K ; partition

$$\mathcal{N} = \mathcal{I}_1 \cup \dots \cup \mathcal{I}_K, \quad \mathcal{I}_k \cap \mathcal{I}_\ell = \emptyset \quad (k \neq \ell),$$

preferably with folds of approximately equal size.

Output.

$$\{\text{CV}(\lambda_m) : m = 1, \dots, M\}, \quad \hat{\lambda}_{\text{CV}}, \quad \hat{\beta}_{\text{CV}}.$$

1. For $m = 1, \dots, M$, set $\lambda \leftarrow \lambda_m$ and repeat over folds.

(a) For $k = 1, \dots, K$, define

$$\mathcal{T}_k \leftarrow \mathcal{N} \setminus \mathcal{I}_k.$$

(b) Fit the Cox lasso on the training data:

$$\hat{\beta}^{-k}(\lambda) \leftarrow \arg \min_{\beta \in \mathbb{R}^p} \{-\ell_{\mathcal{T}_k}(\beta) + \lambda \|\beta\|_1\}.$$

(c) Compute

$$D_{k,m}^{\text{full}} \leftarrow \text{Dev}_{\mathcal{N}}\{\hat{\beta}^{-k}(\lambda)\}, \quad D_{k,m}^{\text{train}} \leftarrow \text{Dev}_{\mathcal{T}_k}\{\hat{\beta}^{-k}(\lambda)\}.$$

(d) Compute the fold contribution using the van Houwelingen subtraction trick:

$$\hat{D}_{k,m}^{\text{CV}} \leftarrow D_{k,m}^{\text{full}} - D_{k,m}^{\text{train}}.$$

2. Aggregate

$$\text{CV}(\lambda_m) \leftarrow \sum_{k=1}^K \hat{D}_{k,m}^{\text{CV}}.$$

3. Select

$$\hat{\lambda}_{\text{CV}} \leftarrow \arg \min_{\lambda_m \in \Lambda} \text{CV}(\lambda_m).$$

4. Refit on the full data:

$$\hat{\beta}_{\text{CV}} \leftarrow \arg \min_{\beta \in \mathbb{R}^p} \{-\ell_{\mathcal{N}}(\beta) + \hat{\lambda}_{\text{CV}} \|\beta\|_1\}.$$

5. Return the CV curve, $\hat{\lambda}_{\text{CV}}$, and $\hat{\beta}_{\text{CV}}$.

3.1 Optional one-standard-error rule

The summed CV score $\text{CV}(\lambda_m)$ and the averaged CV score have the same minimizer. For the one-standard-error rule, use the fold-averaged score

$$\overline{\text{CV}}(\lambda_m) = \frac{1}{K} \sum_{k=1}^K \hat{D}_{k,m}^{\text{CV}}.$$

Define the standard error of the fold-wise CV scores at fixed penalty value λ_m by

$$\widehat{\text{SE}}_{\text{CV}}(\lambda_m) = \frac{\text{sd}(\hat{D}_{1,m}^{\text{CV}}, \dots, \hat{D}_{K,m}^{\text{CV}})}{\sqrt{K}}.$$

Let

$$m_{\min} = \arg \min_m \overline{\text{CV}}(\lambda_m).$$

Then choose the largest penalty whose averaged CV score is within one standard error of the minimum:

$$\hat{\lambda}_{\text{1SE}} = \max \left\{ \lambda_m : \overline{\text{CV}}(\lambda_m) \leq \overline{\text{CV}}(\lambda_{m_{\min}}) + \widehat{\text{SE}}_{\text{CV}}(\lambda_{m_{\min}}) \right\}.$$

After selecting $\hat{\lambda}_{\text{1SE}}$, refit on the full data with that penalty.

3.2 Implementation notes for cross-validation

Note 1. The direct left-out partial likelihood can be unstable when a fold is small because Cox risk sets inside a validation fold can be too small. The subtraction $\text{Dev}_{\mathcal{N}} - \text{Dev}_{\mathcal{T}_k}$ avoids this by computing both terms from sufficiently large risk sets.

Note 2. The direction of subtraction is important: since $\text{Dev}_A(\beta) = -2\ell_A(\beta)$, use $\text{Dev}_{\mathcal{N}}\{\hat{\beta}^{-k}\} - \text{Dev}_{\mathcal{T}_k}\{\hat{\beta}^{-k}\}$, not the reverse.

Note 3. If covariates are standardized inside each training fold, the same fold-specific centering and scaling must be used when evaluating both full-data and training-data deviances. Do not re-standardize the full data separately for $\text{Dev}_{\mathcal{N}}\{\hat{\beta}^{-k}(\lambda)\}$.

Note 4. The summed CV score and averaged CV score differ only by the positive constant K , so they have the same minimizer. The averaged version is convenient for the one-standard-error rule.

Note 5. The standard error $\widehat{\text{SE}}_{\text{CV}}(\lambda_m)$ is the standard error of the fold-wise CV scores at fixed λ_m , not the standard error of λ_m itself.

Note 6. For the one-standard-error rule, folds should preferably have similar sizes. If fold sizes differ substantially, use a fold-size-weighted or per-observation averaged convention consistently.

4 Pre-validation for the Cox lasso

Pre-validation constructs scores $\tilde{\eta}_i^{\text{PV}}$ so that the score for subject i is computed from a model fitted without using subject i 's response (y_i, δ_i) .

Algorithm 5. CoxLassoPreValidation: pre-validation for the Cox lasso

Input. Survival data $\{(y_i, \delta_i, \mathbf{x}_i), i = 1, \dots, N\}$; penalty grid $\Lambda = \{\lambda_1, \dots, \lambda_M\}$; number of outer folds K ; outer-fold partition $\mathcal{N} = \mathcal{I}_1 \cup \dots \cup \mathcal{I}_K$; penalty-selection rule $\mathcal{S}$, for example CV-minimum or one-SE; number of inner folds L used by $\mathcal{S}$; and a pre-specified zero-variance handling rule.

Default zero-variance rule. If a covariate has zero variance inside a training set, remove it from that fold-specific fit and set its fold-specific coefficient to zero.

Output. Pre-validated dataset

$$\mathcal{D}_{\text{PV}} = \{(\tilde{\eta}_i^{\text{PV}}, y_i, \delta_i), i = 1, \dots, N\},$$

fold-specific penalties $\{\hat{\lambda}^{(-k)}\}_{k=1}^K$, standardized-scale coefficients $\{\hat{\beta}^{(-k)}\}_{k=1}^K$, and original-scale affine coefficients $\{(\hat{\alpha}^{(-k)}, \hat{\gamma}^{(-k)})\}_{k=1}^K$.

1. Initialize $\tilde{\eta}_i^{\text{PV}} \leftarrow \text{NA}$ for $i = 1, \dots, N$.

2. For $k = 1, \dots, K$, repeat:

(a) Define $\mathcal{T}_k \leftarrow \mathcal{N} \setminus \mathcal{I}_k$.

- (b) Compute training-fold centering and scaling:

$$\hat{\mu}_j^{(-k)} = \frac{1}{|\mathcal{T}_k|} \sum_{i \in \mathcal{T}_k} x_{ij},$$

$$\hat{\sigma}_j^{(-k)} = \left[\frac{1}{|\mathcal{T}_k|} \sum_{i \in \mathcal{T}_k} (x_{ij} - \hat{\mu}_j^{(-k)})^2 \right]^{1/2}.$$

- (c) Define the fold-specific usable covariate set

$$\mathcal{C}_k = \{j : \hat{\sigma}_j^{(-k)} > 0\}.$$

For $j \notin \mathcal{C}_k$, set $\hat{\beta}_j^{(-k)} = 0$ and $\hat{\gamma}_j^{(-k)} = 0$.

- (d) If $\mathcal{C}_k = \emptyset$, set $\hat{\lambda}^{(-k)} \leftarrow \text{NA}$, $\hat{\alpha}^{(-k)} \leftarrow 0$, and $\hat{\eta}_i^{\text{PV}} \leftarrow 0$ for every $i \in \mathcal{I}_k$. Then skip to the next outer fold.

- (e) Standardize observations using only outer-training-fold parameters:

$$x_{ij}^{(-k)} = \frac{x_{ij} - \hat{\mu}_j^{(-k)}}{\hat{\sigma}_j^{(-k)}}, \quad i \in \mathcal{N}, \quad j \in \mathcal{C}_k.$$

- (f) Let $\ell_{\mathcal{T}_k}^{(-k)}(\beta_{\mathcal{C}_k})$ be the Cox partial log-likelihood on $\mathcal{T}_k$, using standardized covariates indexed by $\mathcal{C}_k$.

- (g) Select the penalty using only the outer training data:

$$\hat{\lambda}^{(-k)} \leftarrow \mathcal{S}(\mathcal{T}_k, \mathcal{C}_k, \Lambda, L).$$

Here $\mathcal{C}_k$ is the outer allowable covariate set. For example, $\mathcal{S}$ may run the Cox lasso cross-validation algorithm above restricted to $\mathcal{T}_k$, using L inner folds. Inside $\mathcal{S}$, any inner-fold standardization, zero-variance filtering, or screening must be recomputed using only the corresponding inner training fold and then applied to the corresponding inner validation fold.

- (h) Fit the Cox lasso on the outer training set:

$$\hat{\beta}_{\mathcal{C}_k}^{(-k)} \leftarrow \arg \min_{\beta_{\mathcal{C}_k} \in \mathbb{R}^{|\mathcal{C}_k|}} \left\{ -\ell_{\mathcal{T}_k}^{(-k)}(\beta_{\mathcal{C}_k}) + \hat{\lambda}^{(-k)} \|\beta_{\mathcal{C}_k}\|_1 \right\}.$$

For $j \notin \mathcal{C}_k$, keep $\hat{\beta}_j^{(-k)} = 0$.

- (i) Convert to the original-scale affine predictor. For $j \in \mathcal{C}_k$, set

$$\hat{\gamma}_j^{(-k)} = \frac{\hat{\beta}_j^{(-k)}}{\hat{\sigma}_j^{(-k)}};$$

for $j \notin \mathcal{C}_k$, set $\hat{\gamma}_j^{(-k)} = 0$. Define

$$\hat{\alpha}^{(-k)} = - \sum_{j \in \mathcal{C}_k} \hat{\mu}_j^{(-k)} \hat{\gamma}_j^{(-k)}.$$

Then

$$(\mathbf{x}_{i, \mathcal{C}_k}^{(-k)})^\top \hat{\beta}_{\mathcal{C}_k}^{(-k)} = \hat{\alpha}^{(-k)} + \mathbf{x}_i^\top \hat{\gamma}^{(-k)}.$$

- (j) For every held-out observation $i \in \mathcal{I}_k$, compute

$$\hat{\eta}_i^{\text{PV}} \leftarrow (\mathbf{x}_{i, \mathcal{C}_k}^{(-k)})^\top \hat{\beta}_{\mathcal{C}_k}^{(-k)},$$

equivalently

$$\hat{\eta}_i^{\text{PV}} \leftarrow \hat{\alpha}^{(-k)} + \mathbf{x}_i^\top \hat{\gamma}^{(-k)}.$$

3. Construct

$$\mathcal{D}_{\text{PV}} \leftarrow \{(\hat{\eta}_i^{\text{PV}}, y_i, \delta_i), i = 1, \dots, N\}.$$

4. Return $\mathcal{D}_{\text{PV}}$, $\{\hat{\lambda}^{(-k)}\}_{k=1}^K$, $\{\hat{\beta}^{(-k)}\}_{k=1}^K$, and $\{(\hat{\alpha}^{(-k)}, \hat{\gamma}^{(-k)})\}_{k=1}^K$.

4.1 Analysis of pre-validated scores

The pre-validated dataset can be analyzed as a one-dimensional survival dataset whose covariate is the pre-validated risk score. One option is to fit

$$\lambda(t | \tilde{\eta}_i^{\text{PV}}) = \lambda_0(t) \exp\{\theta \tilde{\eta}_i^{\text{PV}}\},$$

with partial log-likelihood

$$\ell_{\text{PV}}(\theta) = \sum_{i:\delta_i=1} \left[\theta \tilde{\eta}_i^{\text{PV}} - \log \left\{ \sum_{j:y_j \geq y_i} \exp(\theta \tilde{\eta}_j^{\text{PV}}) \right\} \right].$$

Another option is to split subjects into groups using a threshold c chosen without using survival responses, such as the median of the pre-validated scores, and then display Kaplan-Meier curves or use a log-rank test. Standard errors and p -values should be interpreted cautiously because the scores are cross-fitted and training sets overlap.

4.2 Optional final model for deployment

Pre-validation evaluates a modeling procedure. For deployment, refit the chosen procedure on the full dataset. Compute full-data centering and scaling parameters $\hat{\mu}_j^{\text{full}}$ and $\hat{\sigma}_j^{\text{full}}$, and define

$$\mathcal{C}_{\text{full}} = \{j : \hat{\sigma}_j^{\text{full}} > 0\}.$$

If

$$\mathcal{C}_{\text{full}} = \emptyset,$$

set the deployed linear predictor to zero and skip the full-data Cox lasso fit. Otherwise, select

$$\hat{\lambda}_{\text{full}} \leftarrow \mathcal{S}(\mathcal{N}, \mathcal{C}_{\text{full}}, \Lambda, K),$$

and fit the full-data Cox lasso on $\mathcal{C}_{\text{full}}$. Convert the fitted coefficients to an original-scale affine predictor

$$\hat{\eta}_{\text{new}} = \hat{\alpha}^{\text{full}} + \mathbf{x}_{\text{new}}^{\top} \hat{\gamma}^{\text{full}},$$

where, for $j \in \mathcal{C}_{\text{full}}$,

$$\hat{\gamma}_j^{\text{full}} = \hat{\beta}_j^{\text{full}} / \hat{\sigma}_j^{\text{full}},$$

for $j \notin \mathcal{C}_{\text{full}}$, $\hat{\gamma}_j^{\text{full}} = 0$, and

$$\hat{\alpha}^{\text{full}} = - \sum_{j \in \mathcal{C}_{\text{full}}} \hat{\mu}_j^{\text{full}} \hat{\gamma}_j^{\text{full}}.$$

Equivalently,

$$\hat{\eta}_{\text{new}} = \sum_{j \in \mathcal{C}_{\text{full}}} \frac{x_{\text{new},j} - \hat{\mu}_j^{\text{full}}}{\hat{\sigma}_j^{\text{full}}} \hat{\beta}_j^{\text{full}}.$$

4.3 Implementation notes for pre-validation

Note 1. The score $\tilde{\eta}_i^{\text{PV}}$ for subject i must be computed from a model whose entire fitting process did not use (y_i, δ_i) . This includes variable screening, penalty selection, and any supervised preprocessing.

Note 2. Strictly honest pre-validation selects $\hat{\lambda}^{(-k)}$ inside each outer training set. Using a single $\hat{\lambda}$ chosen from the full dataset leaks outcome information into held-out scores through the tuning parameter.

Note 3. If covariates are standardized, held-out observations in fold k must be standardized using centering and scaling parameters computed from $\mathcal{T}_k$, not from the full dataset.

Note 4. When $\mathcal{S}$ uses inner cross-validation, $\mathcal{C}_k$ is only the outer allowable covariate set. Preprocessing steps inside $\mathcal{S}$, including standardization, zero-variance filtering, and screening, should be recomputed within each inner training fold and applied to the corresponding inner validation fold.

Note 5. If $\hat{\sigma}_j^{(-k)} = 0$, the covariate cannot be standardized in that fold. The default rule removes it from that fold-specific fit and sets its coefficient to zero.

Note 6. Do not compute the original-scale score as $\mathbf{x}_i^\top \hat{\boldsymbol{\gamma}}^{(-k)}$ alone after converting coefficients. The correct original-scale equivalent includes the offset $\hat{\alpha}^{(-k)}$.

Note 7. If tied failure times are present, use the same tie-handling rule in all Cox lasso fits and in the final analysis of $\mathcal{D}_{\text{PV}}$.

References

1. Tibshirani, R. (1997). The lasso method for variable selection in the Cox model. *Statistics in Medicine*, 16(4), 385–395.
2. Hastie, T., Tibshirani, R., and Wainwright, M. (2015). *Statistical Learning with Sparsity: The Lasso and Generalizations*. CRC Press.